

Software Testing Project Requirements

1. Project Overview

The purpose of this project is to provide requirements for testing website features using Selenium and Katalon. You are expected to apply software testing concepts including functional testing, automation testing, API testing, reporting, and test documentation.

You may test any website of your choice. It is highly recommended to use your Graduation Project.

2. Team Structure

The project can be completed individually or in teams of **minimum 4 to maximum 6** students.

3. Required Technologies and Tools

- **Katalon Studio** for automated testing and reporting.
- **Selenium WebDriver** for automation scripts.
- **Postman** for API testing.

4. Functional Requirements

4.1. Katalon Studio (5 Marks)

Only two pages are required in Katalon:

- Sign-up Page (Mandatory)
- One additional functional page other than Login

Acceptance Criteria:

- Successful account creation using valid data.
- Validation messages for invalid or incomplete inputs.
- Successful automation execution in Katalon Studio.
- Generation of execution reports.

4.2. Selenium (10 Marks)

The Login page is mandatory and must be fully tested using Selenium. Additional important website functionalities should also be automated using Selenium.

Mandatory Selenium Pages:

- Login Page (Mandatory)
- Additional functional pages from the website (Not Sign-up)

Acceptance Criteria:

- Successful login with valid credentials.
- Proper handling of invalid login attempts.
- Validation of important workflows within the website.
- Automation scripts should be executed successfully without manual intervention.
- Clear assertions and validations should be included.

4.3. API Testing (5 Marks)

You are required to perform API testing using Postman. It must include a complete End-to-End API testing collection that simulates a realistic workflow.

Requirements

- Create and organize API collections in Postman.
- Use environment variables where applicable.
- Include request validations and assertions.
- Validate response status codes and response bodies.
- Implement positive and negative test scenarios.
- Demonstrate an End-to-End API workflow.

5. Non-Functional Requirements

Reliability

- Automation scripts should be stable and reusable.
- Tests should be executed consistently across multiple runs.
- Proper synchronization and waits should be implemented.
- Code should be organized and maintainable.

Documentation Quality

- All test cases should be clearly documented.
- Bug reports should include sufficient details and screenshots.

6. Required Deliverables (5 Marks)

- Test Plan Document
- Test Suites
- Test Cases Document
- Traceability Matrix
- Bug Reports
- Test Summary Report
- Selenium Automation Scripts
- Katalon Studio Project Files and Reports
- Postman API Collection and Environment Files
- Screenshots from test execution
- Presentation/Demo that demonstrates your work